Assignment 2

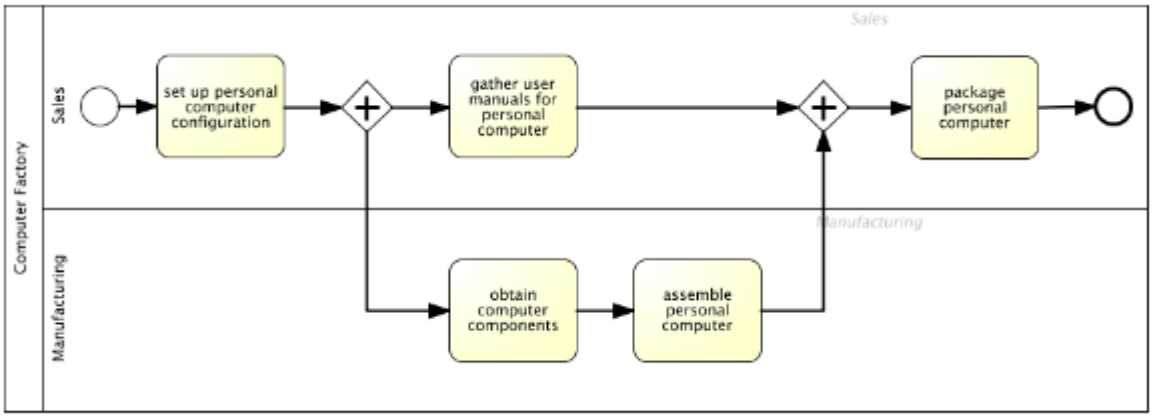
Simulation and Soundness

General Instructions:

- Assignment is individual, any copies will be graded zero.
- If repeated offense, it will deduct grades (-2).
- The file name is your **CourseCode-ID-Assignment2.docx** or **CourseCode-ID-assignment2.pdf**
- Inside the file, start with writing the following:
  - Course Code:
  - Assignment number:
  - ID:
  - Full name (Arabic/English):
  - Email: (the one used on classroom)
- Always use CASE tools or modeling tools for drawing, avoid handwritten diagrams, this can affect your grade.

Part 1: Simulation

The following business process model describes the business process of a computer factory for building personal computers.



Activity processing times	Hourly wages
Set up personal computer configuration: 25 min	Employee in Sales: 30 LE
Gather user manuals for personal computer: 50 min	Employee in Manufacturing: 40 LE
Obtain computer components: 25 min	
Assemble personal computer: 60 min	
Package personal computer: 15 min	

A) How much **time** does **one process instance** take? (1 point) [Write the **equation** used. (1 point)]

a) 2hr 45min

b) 2hr 5 min

c) 1hr 20min

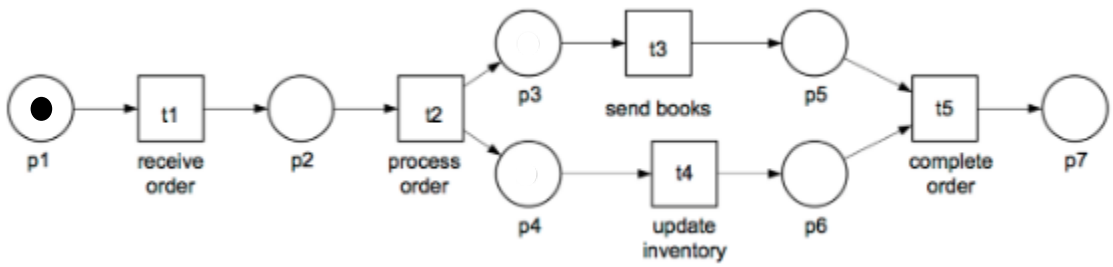
d) 3 hrs

B) What is the **overall cost** of **one process instance**? ***Remark***: The cost calculation of the process differs from the processing time calculation. You need to include all costs with their relative employee cost. (1 point) [Write the **equation** used. (1 point)]

a) 40.00 LE
b) 74.17 LE
c) 101.67 LE
d) 96,67 LE
e) 110.7 LE

Part 2: Soundness

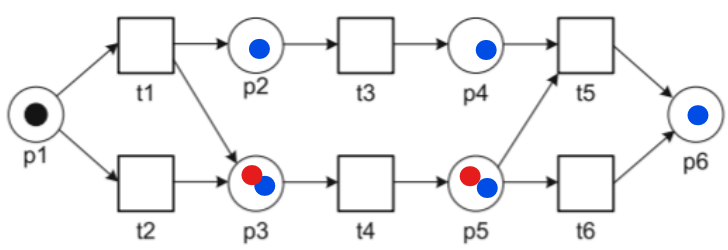
1) Given the following model:



a) Draw the **reachability graph**, **initial marking M0={p1}** (2 points)
b) Based on the answer to the previous question, **the model is:** (1 point)

a) Sound.
b) Not sound. Remaining token in Place P4.
c) Not sound. Remaining token in Place P5.
d) Not sound. Remaining token in Place P6.

2) Given the following model:



a) Draw the **reachability graph** with **initial marking M0={p1}** (2 points)
b) Based on the answer to the previous question, **the model is:** (1 point)

a) Sound.
b) Not sound. Remaining token in Place P4.
c) Not sound. Remaining token in Place P5.
d) Not sound. Remaining token in Place P6.